

Let $\hat{u} = \langle a, b \rangle$.

$$x(s) = x_0 + as, \quad y(s) = y_0 + bs$$

Plug into w , then compute:

$$\frac{dw}{ds}$$

Definition (Directional Derivative):

$$\left. \frac{dw}{ds} \right|_{\hat{u}}$$

Directional derivative represents the slope of the surface in the direction of $\hat{u}$.

By the chain rule:

$$\frac{dw}{ds} = \nabla w \cdot \frac{d\mathbf{r}}{ds}$$

Thus:

$$\left. \frac{dw}{ds} \right|_{\hat{u}} = \nabla w \cdot \hat{u}$$

This is the component of ∇w in the direction of $\hat{u}$.

Example:

$$\left. \frac{dw}{ds} \right|_{\mathbf{i}} = \nabla w \cdot \mathbf{i} = \frac{\partial w}{\partial x}$$

Geometry:

$$\left. \frac{dw}{ds} \right|_{\hat{u}} = \nabla w \cdot \hat{u} = |\nabla w| |\hat{u}| \cos \theta$$

Maximum: occurs when $\cos \theta = 1$

$$\Rightarrow \theta = 0 \Rightarrow \hat{u} = \text{direction of } \nabla w$$

So, the gradient ∇w gives the direction of fastest increase of w .

$$|\nabla w| = \left. \frac{dw}{ds} \right|_{\hat{u}=\text{dir}(\nabla w)}$$

Minimum: occurs when $\cos \theta = -1$

$$\Rightarrow \theta = 180^\circ \Rightarrow \hat{u} \text{ is in direction of } -\nabla w$$

Zero directional derivative:

$$\left. \frac{dw}{ds} \right|_{\hat{u}} = 0 \quad \Rightarrow \quad \hat{u} \perp \nabla w \quad (\theta = 90^\circ)$$

$$\Rightarrow \hat{u} \text{ is tangent to the level curve.}$$

Lagrange Multipliers

We want to find the minimum or maximum of a function

$$f(x, y, z)$$

where the variables are not independent but satisfy a constraint

$$g(x, y, z) = c.$$

Example: Closest point to the origin

Find the point on the hyperbola

$$xy = 3$$

that is closest to the origin.

This is equivalent to minimizing

$$f(x, y) = x^2 + y^2$$

subject to the constraint

$$g(x, y) = xy = 3.$$

Observation: At the minimum, the level curve of f is tangent to the constraint curve $g(x, y) = 3$.

Key Idea

At the point of tangency, the gradients are parallel:

$$\nabla f \parallel \nabla g$$

So,

$$\nabla f = \lambda \nabla g$$

for some scalar λ (Lagrange multiplier).

System to solve

$$\nabla f = \lambda \nabla g, \quad g(x, y) = 3$$

System of Equations

$$\nabla f = \lambda \nabla g$$

$$f_x = \lambda g_x, \quad f_y = \lambda g_y$$

Given:

$$f = x^2 + y^2, \quad g = xy = 3$$

Partial derivatives:

$$f_x = 2x, \quad f_y = 2y$$

$$g_x = y, \quad g_y = x$$

System:

$$2x = \lambda y, \quad 2y = \lambda x, \quad xy = 3$$

Matrix form:

$$\begin{bmatrix} 2 & -\lambda \\ -\lambda & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Determinant:

$$\begin{vmatrix} 2 & -\lambda \\ -\lambda & 2 \end{vmatrix} = 4 - \lambda^2 = 0$$

$$\lambda = \pm 2$$

Solutions:

If $\lambda = 2$:

$$x = y, \quad x^2 = 3 \Rightarrow (x, y) = (\sqrt{3}, \sqrt{3}) \text{ or } (-\sqrt{3}, -\sqrt{3})$$

If $\lambda = -2$:

$$x = -y$$

Explanation of Method

$$-x^2 = 3 \Rightarrow \text{no solutions}$$

Why is this method valid?

At a constrained maximum or minimum, in any direction along the level curve $g = c$, the rate of change of f is zero.

For any unit vector $\hat{u}$ tangent to $g = c$, we must have:

$$\frac{df}{d\hat{u}} = 0$$

$$\nabla f \cdot \hat{u} = 0$$

So any such $\hat{u}$ is perpendicular to ∇f .

Also, we know that ∇g is perpendicular to the level set of g .

Therefore:

$$\nabla f \parallel \nabla g$$

Warning

This method does not tell whether a solution is a minimum or a maximum.

We can use second derivatives to determine this.

To find minimum or maximum:

We compare the value of F at the various solutions of the Lagrange equation.

Advanced Example

We want to build a pyramid with a given triangular base and given volume, and minimize the total surface area.

$$\text{Volume} = \frac{1}{3} \cdot \text{Area}(\text{base}) \cdot \text{height} \quad (\text{fixed})$$

Let the base lie in the plane $z = 0$:

$$P_1(x_1, y_1, 0), \quad P_2(x_2, y_2, 0), \quad P_3(x_3, y_3, 0)$$

Let u_1, u_2, u_3 be the distances from point P to the three sides.

$$\text{Height of faces} = \sqrt{u_i^2 + h^2}$$

$$\text{Side area} = \frac{1}{2}a_1\sqrt{u_1^2 + h^2} + \frac{1}{2}a_2\sqrt{u_2^2 + h^2} + \frac{1}{2}a_3\sqrt{u_3^2 + h^2}$$

Now, cutting the base into three parts:

$$\text{Area}(\text{base}) = \frac{1}{2}a_1u_1 + \frac{1}{2}a_2u_2 + \frac{1}{2}a_3u_3$$

Lagrange Condition

$$\nabla f = \lambda \nabla g$$

$$\frac{\partial f}{\partial u_1} = \frac{1}{2}a_1 \frac{u_1}{\sqrt{u_1^2 + h^2}} = \lambda \frac{\partial g}{\partial u_1}$$

$$\frac{\partial f}{\partial u_2} = \frac{1}{2}a_2 \frac{u_2}{\sqrt{u_2^2 + h^2}} = \lambda \frac{\partial g}{\partial u_2}$$

$$\frac{\partial f}{\partial u_3} = \frac{1}{2}a_3 \frac{u_3}{\sqrt{u_3^2 + h^2}} = \lambda \frac{\partial g}{\partial u_3}$$

$\Rightarrow P$ is the incenter

Lecture 14: Non-independent Variables

Example:

$$f(P, V, T)$$

or

$$f(x, y, z) \quad \text{where} \quad g(x, y, z) = c$$

If

$$g(x, y, z) = c$$

then

$$z = z(x, y)$$

We want:

$$\frac{\partial z}{\partial x}, \quad \frac{\partial z}{\partial y}$$

Example

$$x^2 + y^2 + z^3 = 8 \quad \text{at} \quad (2, 3, 1)$$

Implicit Differentiation Example

Take differential:

$$2x \, dx + 2y \, dy + (y + 3z^2) \, dz = 0$$

At the point $(2, 3, 1)$:

$$4 \, dx + 2 \, dy + 6 \, dz = 0$$

If we view $z = z(x, y)$, then:

$$dz = -\frac{1}{6}(4 \, dx + 2 \, dy)$$

Hence,

$$\frac{\partial z}{\partial x} = -\frac{4}{6} = -\frac{2}{3}, \quad \frac{\partial z}{\partial y} = -\frac{2}{6} = -\frac{1}{3}$$

General Case

If

$$g(x, y, z) = c$$

then

$$dg = g_x \, dx + g_y \, dy + g_z \, dz = 0$$

Solve for dz :

$$dz = -\frac{g_x}{g_z} \, dx - \frac{g_y}{g_z} \, dy$$

Partial Derivatives

For $\frac{\partial z}{\partial x}$ (set y constant, so $dy = 0$):

$$\frac{\partial z}{\partial x} = -\frac{g_x}{g_z}$$

For $\frac{\partial z}{\partial y}$ (set x constant, so $dx = 0$):

$$\frac{\partial z}{\partial y} = -\frac{g_y}{g_z}$$